

Lab4 main

Database Explorer

postgres@localhost

- flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date
 - departure_airport_id integer
 - arrival_airport_id integer
 - departing_gate varchar(5)
 - arriving_gate varchar(50)
 - airline_id integer
 - status varchar(50)
 - actual_departure date
 - actual_arrival date
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 3

1.sql

```
1 create or replace procedure ins_flight (  
2     returning flight_id into p_new_flight_id,  
3  
4     raise notice 'Inserted % in table flights', p_flight_no;  
5  
6 end;  
7  
8 $$;  
9  
10  
11  
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35  
36  
37  
38 alter table flights
```

call ins_flight(3334, 'Kaz', date(now()), date(now()), 13, 1, 1, null, date(now()), date(now()));

Services

Output Result 15

postgres@localhost

del_pass_book.sql

1.sql 125 ms

defence.sql

test.sql

defence.sql

upd_price_flight.sql

reprice.sql

set_disc.sql

1 row

Lab10 > 1.sql

Lab4 main

Database Explorer

postgres@localhost

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 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date
 - departure_airport_id integer
 - arrival_airport_id integer
 - departing_gate varchar(5)
 - arriving_gate varchar(50)
 - airline_id integer
 - status varchar(50)
 - actual_departure date
 - actual_arrival date
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 3

1.sql

2.sql

```
1 create or replace procedure upd_status_flight(  
2     in p_new_status varchar,  
3     out p_upd_status_id int  
4 )  
5  
6 language plpgsql  
7 as $$  
8 begin  
9     update flights  
10    set status = p_new_status  
11    where flight_id = p_flight_id  
12    returning flight_id, status into p_upd_status_id;  
13  
14    raise notice 'Updated % in table flights', p_upd_status_id;  
15  
16 end;  
17  
18 $$;  
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98  
99  
100
```

call upd_status_flight(13, 'Everything OK', null);

Services

Output Result 2

postgres@localhost

upd_price_flight.sql

reprice.sql

set_disc.sql

new_pass.sql

old_to_new_f.sql

upd_bag_wg.sql

2.sql 367 ms

upd_pass.sql

reschedule.sql

1 row

Lab10 > 2.sql

Lab4 main

Database Explorer

postgres@localhost

- flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date
 - departure_airport_id integer
 - arrival_airport_id integer
 - departing_gate varchar(50)
 - arriving_gate varchar(50)
 - airline_id integer
 - status varchar(50)
 - actual_departure date
 - actual_arrival date
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 3

1 create or replace procedure ret_airport(
5 language plpgsql
6 as \$\$
7 begin
8 open ref_cur for
9 select flights.flight_id, flights.flight_no from flights
10 where flights.departure_airport_id = p_dep_airport_id;
11 end;
12 \$\$;
13
14 begin;
15
16 call ret_airport(
17 p_dep_airport_id 1,
18 ref_cur 'curr_curs'
19);
20
21 fetch all from curr_curs;
22
23 end;

Services

Output

Result 8-2

flight_id	flight_no
1	15 ML-2
2	40 TW-TTT
3	57 CN-51
4	61 AU-WA
5	103 SL-N
6	135 US-OK

49 rows

Lab4 main

Database Explorer

postgres@localhost

- flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date
 - departure_airport_id integer
 - arrival_airport_id integer
 - departing_gate varchar(50)
 - arriving_gate varchar(50)
 - airline_id integer
 - status varchar(50)
 - actual_departure date
 - actual_arrival date
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 3

1 create or replace function avg_delay(
2 p_arr_airport_id int
3)
4 returns int
5 language plpgsql
6 as \$\$
7 declare v_avg_delay int;
8 begin
9 select avg(scheduled_arrival - scheduled_departure)
10 into v_avg_delay
11 from flights
12 where arrival_airport_id = p_arr_airport_id;
13
14 return abs(v_avg_delay);
15 end;
16 \$\$;
17
18 select avg_delay(p_arr_airport_id 1)

Services

Output

avg_delay(1):integer

avg_delay
12

1 row

Lab4 main

Database Explorer

postgres@localhost

- baggage_check
- boarding_pass
- booking
 - booking_flight
 - columns 5
 - booking_flight_id integer
 - booking_id integer
 - flight_id integer
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 2
 - indexes 2
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date

1 create or replace procedure list_pass(
2 in p_flight_num varchar,
3 inout ref_cur refcursor
4)
5 language plpgsql
6 as \$\$
7 begin
8 open ref_cur for
9 select passengers.first_name || ' ' || passengers.last_name as full_name, passengers.passenger_id, f.flight_id
10 join public.booking b 1<->1.n on passengers.passenger_id = b.passenger_id
11 join public.booking_flight bf 1<->1.n on b.booking_id = bf.booking_id
12 join public.flights f 1.n<->1 on f.flight_id = bf.flight_id
13 where flight_no = p_flight_num;
14 end;
15 \$\$;
16
17 begin;
18 call list_pass(p_flight_num 'NP-BA', ref_cur 'curs');
19 fetch all from curs;
20 end;
21

Files

Lab10 ~/Desktop/University/IT

- 1.sql [postgres@localhost]
- 2.sql [postgres@localhost]
- 3.sql [postgres@localhost]
- 4.sql [postgres@localhost]
- 5.sql [postgres@localhost]

Scratches and Consoles

Services

Output Result 3-2

full_name	passenger_id	flight_no	booking_flight_id
Muhammad Faw	19	NP-BA	1
Leroi O'Shevl	132	NP-BA	536

2 rows

Lab10 5.sql

20:5 LF UTF-8 4 spaces

Lab4 main

Database Explorer

postgres@localhost

- baggage_check
- boarding_pass
- booking
 - booking_flight
 - columns 5
 - booking_flight_id integer
 - booking_id integer
 - flight_id integer
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 2
 - indexes 2
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date

3 create or replace procedure greatest_fl(
4 begin
5
6 select p.first_name || ' ' || p.last_name as full_name, p.passenger_id, count(bf.flight_id) as num_of_f
7 into p_full_name, p_pass_id, num_of_flights
8 from passengers p
9 join public.booking b 1<->1.n on p.passenger_id = b.passenger_id
10 join public.booking_flight bf 1<->1.n on b.booking_id = bf.booking_id
11 group by p.passenger_id
12 order by num_of_fl DESC
13 limit 1;
14 end;
15 \$\$;
16
17 call greatest_fl(null, null, null);
18
19
20
21
22
23
24
25

Files

Lab10 ~/Desktop/University/IT

- 1.sql [postgres@localhost]
- 2.sql [postgres@localhost]
- 3.sql [postgres@localhost]
- 4.sql [postgres@localhost]
- 5.sql [postgres@localhost]
- 6.sql [postgres@localhost]

Scratches and Consoles

Services

Output Result 15

p_full_name	p_pass_id	num_of_flights
Sheela Roux	68	18

1 row

Lab10 6.sql

22:34 LF UTF-8 4 spaces

Lab4 main

Database Explorer

- postgres@localhost
 - baggage_check
 - boarding_pass
 - booking
 - booking_flight
 - columns 5
 - booking_flight_id integer
 - booking_id integer
 - flight_id integer
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 2
 - indexes 2
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date

```

1 create or replace procedure more_tf(
2   ref_cur refcursor
3 )
4 language plpgsql
5 as $$
6 begin
7   open ref_cur for
8     select flights.flight_no, flights.flight_id, (actual_arrival - actual_departure) as delay from flights
9     where abs(actual_arrival - actual_departure) >= 24;
10 end;
11 $$;
12
13 begin;
14 call more_tf(ref_cur 'cur_ref');
15 fetch all from cur_ref;
16 end;
17

```

Files

- Lab10 ~/Desktop/University/
 - 1.sql (postgres@localhost)
 - 2.sql (postgres@localhost)
 - 3.sql (postgres@localhost)
 - 4.sql (postgres@localhost)
 - 5.sql (postgres@localhost)
 - 6.sql (postgres@localhost)
 - 7.sql (postgres@localhost)
- Scratches and Consoles

Services

Output Result 2-3

flight_no	flight_id	delay
177 CA-DN	206	224
178 BS-NP	207	59
179 NZ-CAN	208	186
180 PG-ESW	209	245
181 US-AK	210	186
182 CN-32	211	186

Lab10 7.sql 13:1 (60 chars, 3 line breaks) LF UTF-8 4 spaces

Lab4 main

Database Explorer

- postgres@localhost
 - baggage_check
 - boarding_pass
 - booking
 - booking_flight
 - columns 5
 - booking_flight_id integer
 - booking_id integer
 - flight_id integer
 - created_at date
 - update_at date
 - keys 2
 - foreign keys 2
 - indexes 2
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date

```

1 create or replace function conter_air()
2 returns table (
3   airline_id int,
4   counter int
5 )
6 language sql
7 as $$
8   select airline.airline_id, count(f.flight_id) as counter
9   from airline
10  left join public.flights f on airline.airline_id = f.airline_id
11  group by airline.airline_id;
12
13
14
15
16 select * from conter_air();
17

```

Files

- Lab10 ~/Desktop/University/
 - 1.sql (postgres@localhost)
 - 2.sql (postgres@localhost)
 - 3.sql (postgres@localhost)
 - 4.sql (postgres@localhost)
 - 5.sql (postgres@localhost)
 - 6.sql (postgres@localhost)
 - 7.sql (postgres@localhost)
 - 8.sql (postgres@localhost)
- Scratches and Consoles

Services

Output Result 7

airline_id	counter
1	42
2	29
3	4
4	34
5	41
6	40

Lab10 8.sql 16:28 LF UTF-8 4 spaces

Lab4 main

Database Explorer

- postgres@localhost
 - booking_flight
 - columns 5
 - keys 2
 - foreign keys 2
 - indexes 2
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date
 - departure_airport_id integer
 - arrival_airport_id integer
 - departing_gate varchar(50)
 - arriving_gate varchar(50)
 - airline_id integer
 - status varchar(50)
 - actual_departure date
 - actual_arrival date

```
1 create or replace procedure ticket_avg(  
2   in p_flight_id int,  
3   out avg_price int  
4 )  
5 language plpgsql  
6 as $$  
7 begin  
8   select avg(booking.price) into avg_price from booking  
9   join public.booking_flight bf 1<->1.n on booking.booking_id = bf.booking_id  
10  join public.flights f 1.n<->1 on bf.flight_id = f.flight_id  
11  where bf.flight_id = p_flight_id;  
12 end;  
13 $$;  
14  
15 call ticket_avg(1, null)
```

Files

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 - 5.sql (postgres@localhost)
 - 6.sql (postgres@localhost)
 - 7.sql (postgres@localhost)
 - 8.sql (postgres@localhost)
 - 9.sql (postgres@localhost)
- Scratches and Consoles

Services

Output Result 2

avg_price
4950

1 row

Lab4 main

Database Explorer

- postgres@localhost
 - booking_flight
 - columns 5
 - keys 2
 - foreign keys 2
 - indexes 2
 - flights
 - columns 14
 - flight_id integer
 - flight_no varchar(50)
 - scheduled_departure date
 - scheduled_arrival date
 - departure_airport_id integer
 - arrival_airport_id integer
 - departing_gate varchar(50)
 - arriving_gate varchar(50)
 - airline_id integer
 - status varchar(50)
 - actual_departure date
 - actual_arrival date

```
1 create or replace procedure max_price(  
2   out p_flight_no varchar,  
3   out p_dep int,  
4   out p_arr int,  
5   out p_price int  
6 )  
7 language plpgsql  
8 as $$  
9 begin  
10  select f.flight_no, f.departure_airport_id, f.arrival_airport_id, b.price  
11  into p_flight_no, p_dep, p_arr, p_price  
12  from booking b  
13  join public.booking_flight bf 1<->1.n on b.booking_id = bf.booking_id  
14  join public.flights f 1.n<->1 on bf.flight_id = f.flight_id  
15  order by price desc  
16  limit 1;  
17 end;  
18 $$;  
19  
20 call max_price(null, null, null, null)
```

Files

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 - 4.sql (postgres@localhost)
 - 5.sql (postgres@localhost)
 - 6.sql (postgres@localhost)
 - 7.sql (postgres@localhost)
 - 8.sql (postgres@localhost)
 - 9.sql (postgres@localhost)
 - 10.sql (postgres@localhost)
- Scratches and Consoles

Services

Output Result 2

p_flight_no	p_dep	p_arr	p_price
NP-BA	19	8	4950

1 row